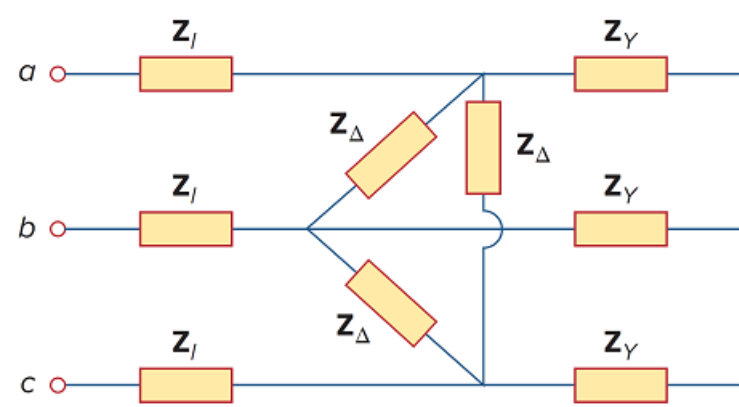


Problem

The circuit in Fig. 12.48 is excited by a balanced three-phase source with a line voltage of 210 V. If $\mathbf{Z}_I = 1 + j1 \, \Omega$, $\mathbf{Z}_\Delta = 24 - j30 \, \Omega$, and $\mathbf{Z}_Y = 12 + j5 \, \Omega$, determine the magnitude of the line current of the combined loads.

Figure 12.48



Step-by-step solution

Step 1 of 2

Convert the delta load, $\mathbf{Z}_\Delta$, to IES equivalent WYE load:

$$\mathbf{Z}_{ye} = \frac{\mathbf{Z}_\Delta}{3} = 8 - j10$$

$$\mathbf{Z}_p = \mathbf{Z}_Y \parallel \mathbf{Z}_{ye} = \frac{(12 + j5)(8 - j10)}{20 - j5} = 8.076 \angle -14.68^\circ$$

$$\mathbf{Z}_p = 7.812 - j2.047 \quad \mathbf{Z}_T = \mathbf{Z}_p + \mathbf{Z}_I = 8.812 - j1.047$$

$$\mathbf{Z}_T = 8.874 \angle -6.78^\circ$$

Step 2 of 2

We now use per phase equivalent circuit,

$$\begin{aligned} I_a &= \frac{V_p}{\mathbf{Z}_p + \mathbf{Z}_I} \\ \Rightarrow I_a &= \frac{210}{\sqrt{3}(8.874 \angle -6.78^\circ)} = 13.66 \angle 6.78^\circ \end{aligned}$$

Where,

$$V_p = \frac{210}{\sqrt{3}}$$

$$I_L = |I_a| = 13.66 \text{ A}$$